第九讲

刚架曲杆内力

作业： 5-7(b,c),8(d),9,17, 18,21.

第五章　弯曲内力

5.1 梁的约束与类型

5.2 剪力弯矩图

5.3 剪力弯矩与载荷集度间的微分关系

5.4 刚架和曲杆的内力

5.5 特殊问题分析

5.6 降低最大弯矩的措施

很重要　挺简单

守规矩　会检查

第五章 弯曲内力

5.4 刚架和曲杆的内力

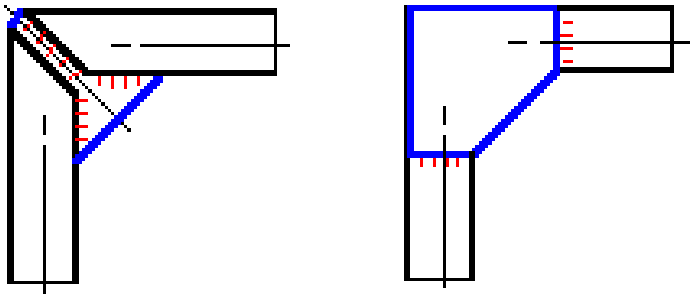
1、刚架内力

2、曲杆内力

1、刚架内力

刚架

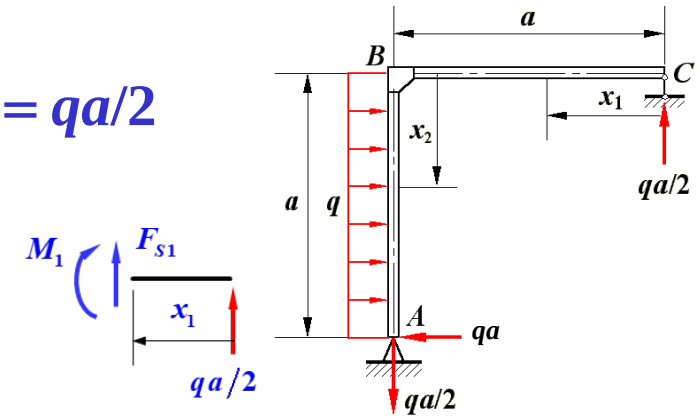
- 用刚性接头连接的杆系结构。
- 接头两侧截面的位移、转角相同；
- 可传力、力矩。



1、刚架内力

整体平衡 ➡ $F_{Ax}=qa,\quad F_{Cy}=F_{Ay}=qa/2$

分离体平衡 ➡ 截面内力



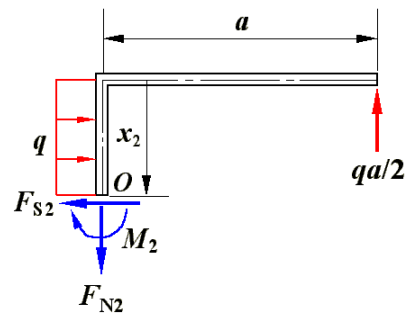
BC 段：

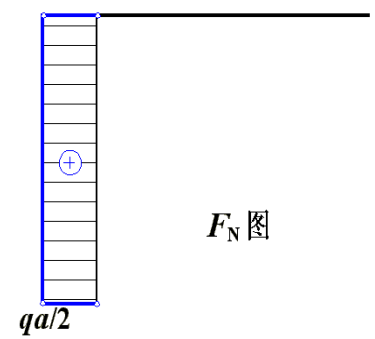
$$F_{N1}=0,\quad F_{S1}=-\frac{qa}{2},\quad M_1=\frac{qa}{2}x_1$$

AB 段：

$$F_{N2}=\frac{qa}{2},\quad F_{S2}=qx_2,$$

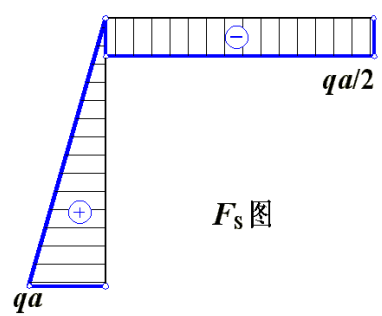
$$M_2=\frac{qa}{2}a-\frac{q}{2}x_2^2$$

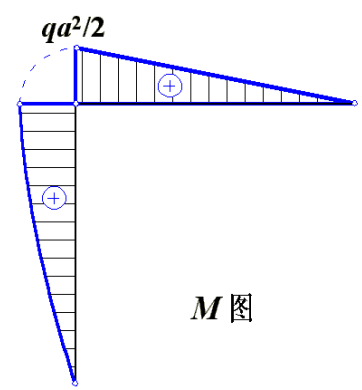




弯矩符号规定与水平段一致

弯矩画在受压边

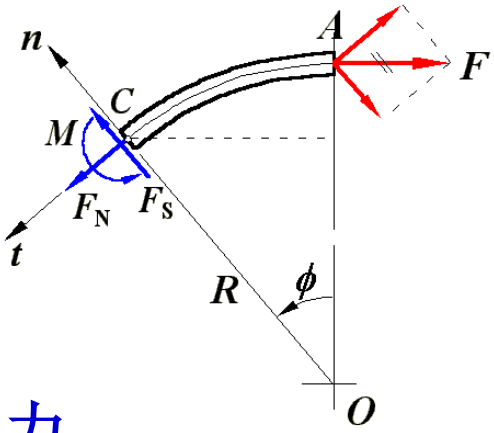


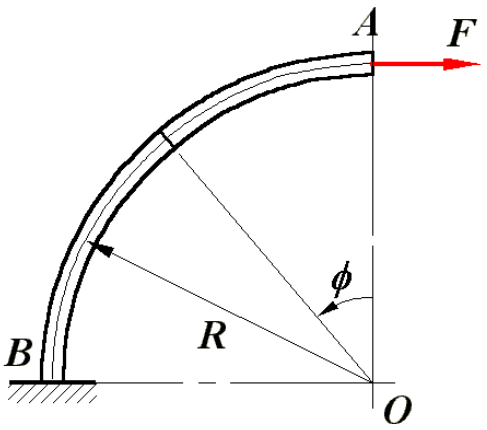


2、曲杆内力

曲杆

- 轴线为曲线的杆件；
- 常见圆弧形曲杆。





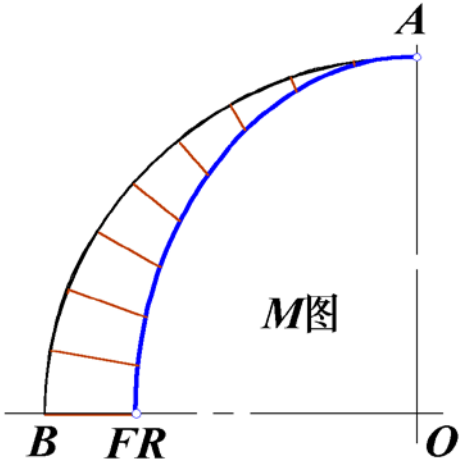
分离体平衡 ➡ 截面内力

$$F_S = F\sin\phi$$

$$F_N = F\cos\phi$$

$$M = FR(1-\cos\phi)$$

弯矩符号：
使曲率增大为正



弯矩画在受压边

第五章 弯曲内力

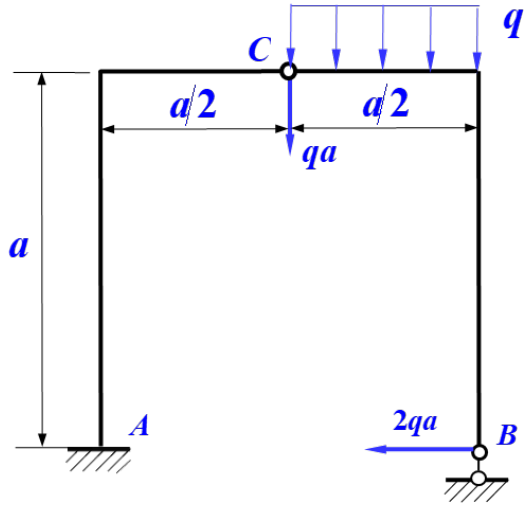
5.5 特殊问题分析

1、梁间铰链受力分析

2、受任意分布载荷直梁的剪力弯矩

3、受均布载荷曲杆内力分析

1、组合结构受力分析

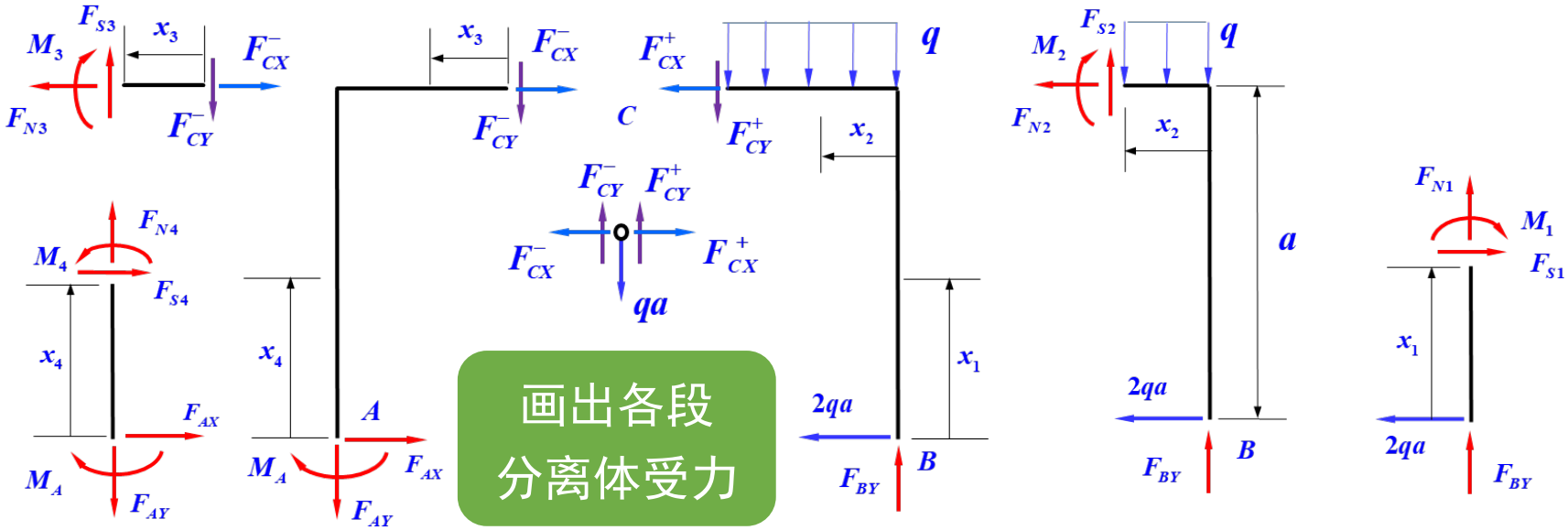


拆开铰链

画好受力分析图

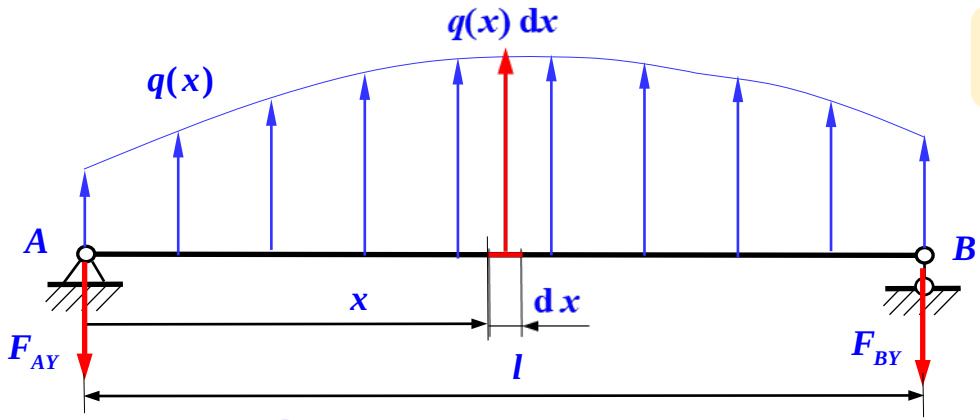
各部分平衡 → 约束反力

分离体平衡 → 截面内力



画出各段
分离体受力

2、受任意分布载荷直梁的剪力弯矩

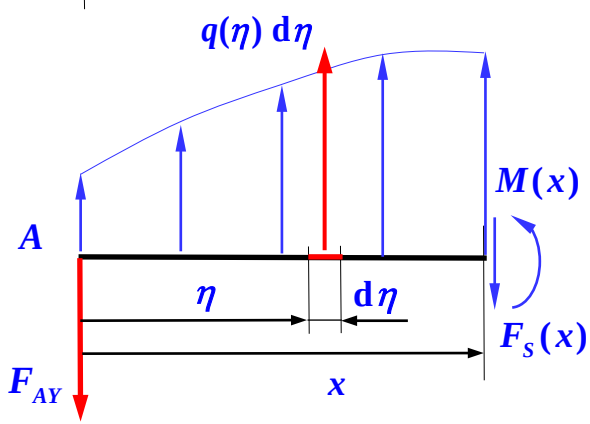


整体平衡

$$F_{AY}=\frac{1}{l}\int_0^l (l-x)q(x)\,\mathrm{d}x$$

$$F_{BY}=\frac{1}{l}\int_0^l xq(x)\,\mathrm{d}x$$

验证 $$\boxed{F_{AY}+F_{BY}=\int_0^l q(x)\,\mathrm{d}x}$$



分离体平衡

载荷大小改变

取微段dη，分析载荷对截面内力的贡献

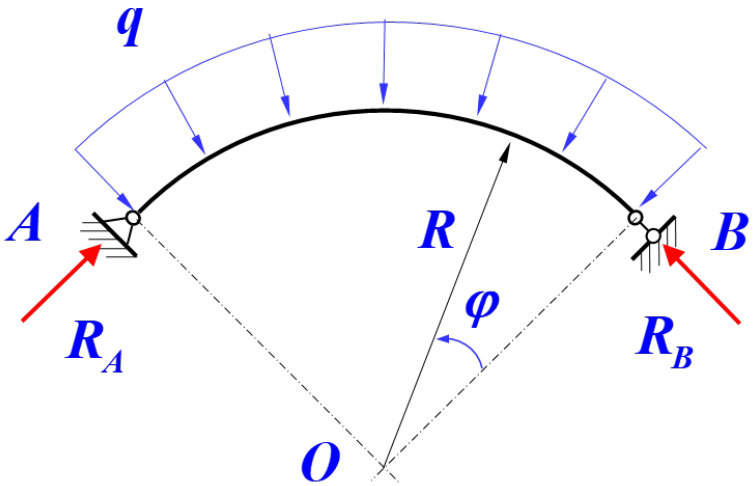
$$\boxed{\begin{aligned}\mathrm{d}F_S&=q(\eta)\,\mathrm{d}\eta\\ \mathrm{d}M&=(x-\eta)q(\eta)\,\mathrm{d}\eta\end{aligned}}$$

$$F_S(x)=\int \mathrm{d}F_S-F_{AY}=\int_0^x q(\eta)\,\mathrm{d}\eta-F_{AY}$$

$$M(x)=\int \mathrm{d}M-F_{AY}x=\int_0^x (x-\eta)q(\eta)\,\mathrm{d}\eta-F_{AY}x$$

3、受均布载荷曲杆内力分析

1/4圆弧小曲率杆，受法向均布载荷作用，求内力。



整体平衡 ➡ $R_A = R_B = qR$

分离体平衡　载荷方向改变

取微段Rdα，分析载荷对截面内力的贡献

$$F_N(\varphi) = \int \mathrm{d}F_N - qR\cos\varphi$$

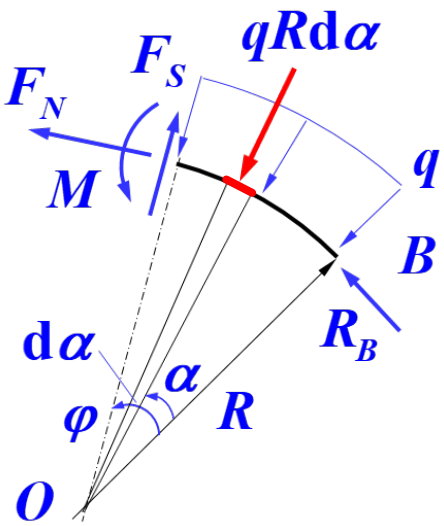
$$F_S(\varphi) = \int \mathrm{d}F_S - qR\sin\varphi$$

$$M(\varphi) = \int \mathrm{d}M - qR^2(1-\cos\varphi)$$

$$\mathrm{d}F_N = -qR\,\mathrm{d}\alpha\sin(\varphi-\alpha)$$

$$\mathrm{d}F_S = qR\,\mathrm{d}\alpha\cos(\varphi-\alpha)$$

$$\mathrm{d}M = qR\,\mathrm{d}\alpha R\sin(\varphi-\alpha)$$



$$F_N(\varphi) = -qR,$$

$$F_S(\varphi) = 0,\ M(\varphi) = 0$$

圆拱法向受压

只有轴压无弯剪

第五章 弯曲内力

5.6 降低最大弯矩的措施

1、移动支座的等弯矩设计

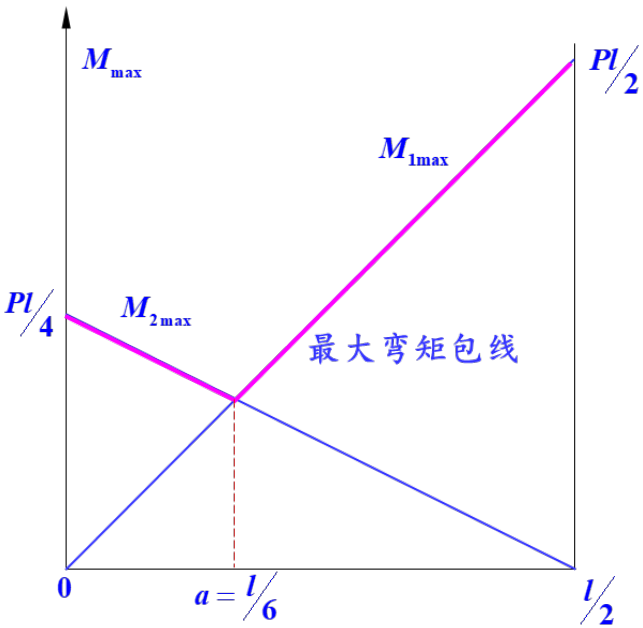
2、加配重降低最大弯矩

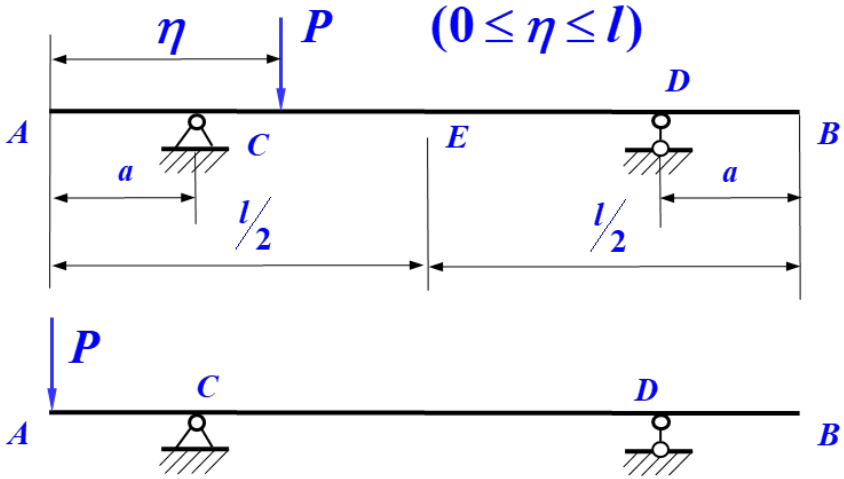
1、移动支座的等弯矩设计

外伸梁受移动载荷作用。

为使最大弯矩最小，a=?

$$M_{1\max} = Pa,\ M_{2\max} = \frac{P}{4}(l-2a)$$





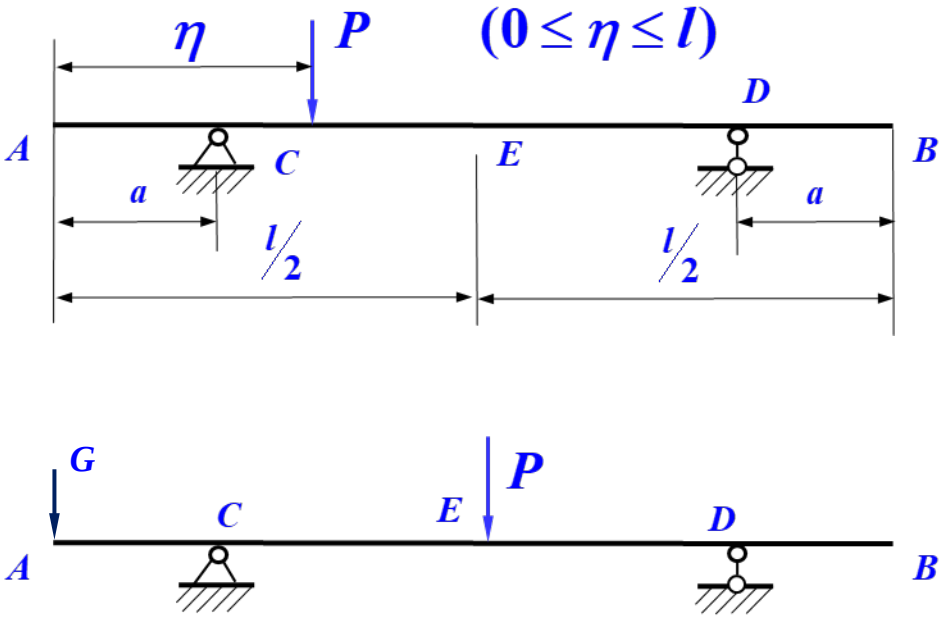
$M_{1\max} = M_{2\max}$

等弯矩　最大弯矩最小

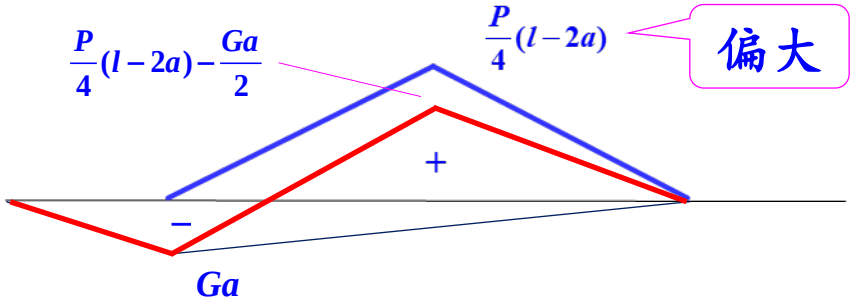
2、加配重降低最大弯矩

熊猫（重P）猴子（重G）过桥AB，$l > 6a$。经试探，猴子轻，可安全通过；熊猫重，可通过AC段，但CD段难以通过。

问：有什么办法?



加配重
降低最大弯矩



本讲：剪力弯矩图

很重要，挺简单

守规矩，会检查

下讲：弯曲正应力